



(Academic Session: 2024 - 2025)

ONLINE CLASSROOM CONTACT PROGRAMME

JEE (MAIN + ADVANCED)

IOQM MOCK TEST # 02

READ THE INSTRUCTIONS CAREFULLY

Important Instructions:

1. This paper contains a total number of 30 questions.
2. Duration of the test is 3 hours.

Your Hard Work Leads to Strong Foundation

ALLEN Digital



Corporate Address : One Biz Square, A-12 (a), Road No. 1, Indraprastha Industrial Area, Kota - 324005 (Raj.)



+91-9513736499



+91-7849901001



wecare@allen.in



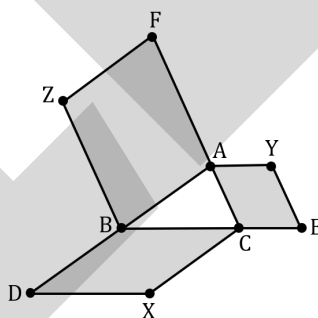
www.allen.in

SECTION-A (Maximum Marks:20)

- **Section-A** contains 10 questions carry 2 marks each & there are no negative marking.

- How many triples of positive integers (a, b, c) are there such that $a! + b^3 = 18 + c^3$?
- Find the minimum number of perfect cubes such that their sum is equal to 346^{346} .
- Find number of non-negative values of x for which $\sqrt[3]{13 + \sqrt{x}} + \sqrt[3]{13 - \sqrt{x}}$ is an integer.
- Find the number of all integers $n > 1$, for which the number $a^{25} - a$ is divisible by n for every integer a .
- For any positive integer n , let

$$f(n) = \frac{4n + \sqrt{4n^2 - 1}}{\sqrt{2n+1} + \sqrt{2n-1}}$$
 Evaluate the sum $f(1) + f(2) + \dots + f(12)$
- Let D be a point inside $\triangle ABC$ such that AD bisects $\angle A$ and $AD \perp BD$. Let M be the midpoint of BC . If $AB = 441$ and $AC = 289$ then find MD .
- Refer to the diagram on the below. Given $\triangle ABC$, extend AB to D such that $AB = BD$, extend BC to E such that $BC = 2CE$ and extend CA to F such that $AF = 2AC$. Draw parallelograms $BCXD$, $ACEY$ and $ABZF$. If the total area of these three parallelograms is 175 cm^2 , find the area of $\triangle ABC$ in cm^2 .



- Let a, b, c, d be digits such that $abcd$ be a four digit number satisfying the property $abcd \times 4 = dcba$ where $dcba$ is also a four digit number. Evaluate : $a + b + c + d$.
- Find the greatest two digit prime number that divides $\frac{96!}{(48!)^2} - 1$.
- Determine the number of ordered pairs of integers (m, n) for which $mn \geq 0$ and $m^3 + n^3 + 99mn = 33^3$.

SECTION-B : (Maximum Marks: 30)

- **Section-B** contains 10 questions carry 3 marks each & there are no negative marking.

- Define a function $g: \mathbb{N} \rightarrow \mathbb{R}$ such that $g(x) = \sqrt{4^x + \sqrt{4^{x+1} + \sqrt{4^{x+2} + \dots}}}$. Find the sum of last 4 digits in the decimal representation of $g(2024)$.
- In the isosceles triangle ABC the angle BAC is a right angle. Point D lies on the side BC and satisfies $BD = 2 \times CD$. Point E is the foot of the perpendicular of the point B on the line AD. Find the angle CED.
- There exists exactly one $n \in \mathbb{N}$, such that $n^4 + 6n^3 + 11n^2 + 3n + 31$ is a perfect square. Find that 'n'.
- Consider the sequence $a_n = 2 - \frac{1}{n^2 + \sqrt{n^4 + \frac{1}{4}}}$, $n \geq 1$
Evaluate: $\frac{1}{2} [\sqrt{a_1} + \sqrt{a_2} + \dots + \sqrt{a_{119}}]$
- In a non-degenerate triangle ABC, $\angle C = 3\angle A$; $BC = 27$; $AB = 48$; Find the length of side AC.
- Consider the sequence of numbers $\left[n + \sqrt{n} + \frac{1}{2} \right]$ for $n \geq 1$, where $[x]$ denotes the greatest integer not exceeding x. Find the largest two digit number which is missing in the sequence.
- ABCD is a square, and P is a point inside with $|PD| = 1$, $|PA| = 2$, $|PB| = 3$.
Find $\angle APD$.
If $\angle APB = \theta$ (degrees) then evaluate $\left[\frac{\theta}{3} \right]$ where $[x]$ denotes the greatest integer less than or equals to x.
- A point P inside the equilateral triangle ABC of side s has distances 3, 4, and 5 from the vertices A, C and B, respectively. If $s^2 = a + b\sqrt{c}$, where $a, b, c \in \mathbb{N}$ and $\sqrt{c} \notin \mathbb{Q}$ then find the least possible sum $a + b + c$.
- For any positive integer n, let $\langle n \rangle$ denote the closest integer to $\sqrt{n}$. Evaluate $\sum_{n=1}^{\infty} \frac{2^{\langle n \rangle} + 2^{-\langle n \rangle}}{2^n}$
- Find the greatest integer less than or equals to $\frac{\sum_{n=1}^{99} \sqrt{10 + \sqrt{n}}}{\sum_{n=1}^{99} \sqrt{10 - \sqrt{n}}}$

- **Section-C** contains 10 questions carry 5 marks each & there are no negative marking.

- The side length of the smallest equilateral triangle in which three disks with radii 2, 3 and 4 units without common inner points can be placed is $a\sqrt{b}$ units where a, b are natural numbers and $\sqrt{b}$ is an irrational number. Evaluate : $a + b$.
- On the sides of a non-obtuse triangle ABC a square, a regular n -gon and a regular m -gon ($n, m > 5$) are constructed externally, so that their centres are vertices of a regular triangle. Find the largest angle of $\triangle ABC$.
- In a competition 8 judges marked the contestants by yes or no. It is known that for any two contestants, two judge gave both a yes; two judges gave the first one a yes and the second one a no, two judges gave the first one a no and the second one a yes, and finally, two judges gave both a no. What is the greatest possible number of contestants?
- Let $S = \{1, 2, \dots, 2005\}$. If there is at least one prime number in any subset of S consisting of n pairwise coprime numbers, find the minimum value of n .
- Let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{7, 8, 9, \dots, n\}$. For each i , choose three numbers in A and two numbers in B , and form a set A_i ($i = 1, 2, \dots, 20$), with five elements, such that $|A_i \cap A_j| \leq 2$, $1 \leq i < j \leq 20$, find the minimum value of n .
- Sharavari shoots free throws on a basketball court. She hits the first and misses the second, and thereafter the probability that she hits the next shot is equal to the proportion of shots she has hit so far. If p is the probability she hits exactly 50 of her first 100 shots, then evaluate $\frac{1}{p}$.
- A number n has sum of digits 100, whilst $44n$ has sum of digits 800. Let 's' the sum of the digits of $3n$, then find the largest two digit divisor of s .
- For some positive integers p, q, r we have $6^p + 2^q + 2 = r^2$. Find the value of $p + q + r$.
- For each positive integer n , let a_n denote the product of all digits of n . If n is a single digit number, then $n = a_n$. Find the sum of all possible positive integers n such that $n^2 - 17n + 56 = a_n$.
- A natural number of five digits is called PANDA if it satisfies the following conditions
 - All its digits are different
 - The digit on the far left is equal to the sum of the other four digits. Example, 91350 is a PANDA number since $9 = 1 + 3 + 5 + 0$ but 54210 is not since $5 \neq 4 + 2 + 1 + 0$.
 - The number is less than 90,000.
 Find how many PANDA numbers exist.